NAMING COMPOUNDS 15

If a compound contains only two elements its name usually ends in '-ide'. In a compound that contains a metal and a non-metal, the metal comes first in the compound's name. For example, iron (III) oxide, Fe_2O_3, contains the metal iron, Fe, and the non-metal oxygen, O. Sodium chloride (NaCl) contains the metal sodium, Na, and the non-metal chlorine, Cl.

These are some name changes for elements present in compounds:

F, fluorine → fluoride I, iodine → iodide Br, bromine → bromide

O, oxygen → oxide N, nitrogen → nitride S, sulfur → sulfide

1 Name two things that the elements above have in common.

2 Copy and complete the table. The first one has been done for you.

Compound	Elements present	Symbols for elements
magnesium oxide	magnesium and oxygen	Mg, O
copper chloride		
zinc oxide		
iron sulfide		
sodium iodide		
magnesium nitride		
potassium bromide		

Often in chemistry atoms form groups that stay together during reactions, acting in a similar way to a single atom. For example, when four oxygen atoms bond with a sulfur atom, a sulfate ion, SO_4^{2-}, is formed. Now consider magnesium sulfate ($MgSO_4$) in a reaction. This compound can 'break up' to form a magnesium ion (Mg^{2+}) and a sulfate 'unit' with the five atoms that form the sulfate unit remaining intact. A group of atoms that stays together in chemical reactions is called a **radical**. If more than two elements are present in a compound and one of these is oxygen, then the compound's name usually ends in '-ate'. These radicals (groups of atoms) are common parts of compounds:

$-SO_4$ = sulfate $-CO_3$ = carbonate $-NO_3$= nitrate $-PO_4$ = phosphate

Naming compounds that contain radicals is not as confusing as it may first appear. The metals keep most (or a recognisable part) of their names and the presence of oxygen in the radical group usually denotes an '-ate' ending to the name.

3 Copy and complete the following table. The first one has been done for you.

Formula	Elements present	Name of compound
$CaCO_3$	calcium, carbon, oxygen	calcium carbonate
$CuSO_4$		
$FePO_4$		
$AgNO_3$		
$CuCO_3$		
$BaSO_4$		
$NaNO_3$		

4 What elements are found in the hydroxide, -OH, radical?

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Some chemicals have 'common' names. NH_3 is usually called *ammonia* (rather than nitrogen hydride) and H_2O is referred to as *water* (not hydrogen oxide). *Vinegar* is a dilute solution of ethanoic (acetic) acid and has the formula CH_3COOH.

5 Name the elements present in:

a ammonia _____

b water _____

c vinegar _____

Working with formulae

The formula of a compound tells the type and ratios of atoms present. Water has the formula H_2O, which means there are two atoms of hydrogen and one atom of oxygen present in the simplest unit of water. Calcium sulfate has the formula $CaSO_4$ and so has one calcium atom, one sulfur atom and four oxygen atoms in one calcium sulfate particle.

If there are brackets in the formula with a number subscript outside, then the atoms in the brackets are multiplied by that number. For example, magnesium hydroxide, $Mg(OH)_2$, has one magnesium atom, *two* oxygen atoms and *two* hydrogen atoms. The brackets show that the hydroxide unit is a radical. Two hydroxide units bond with one magnesium atom to form magnesium hydroxide. Magnesium has a valence of +2 and each hydroxide unit has a valence of –1, so two hydroxides are required to create an electrically-neutral compound.

6 Write the name of each compound and give the names and numbers of atoms present in one unit of that compound.

a MnO_2 _____

b $CoCl_2$ _____

c $MgSO_4$ _____

d $AlCl_3$ _____

e $Zn(OH)_2$ _____

f $CuCO_3$ _____

g Al_2O_3 _____

h $Ba(NO_3)_2$ _____

i K_3PO_4 _____

Valency

The valency of an element in a covalent compound is the number of bonds formed by one atom of the element in one molecule of the compound. The usual valency of an element is either its group number (given in the Periodic Table) or, if this is greater than 4, 8 minus the group number. The valency of individual ions in ionic compounds is the charge on that ion.

Thus the valency of Group I elements is 1; the valencies of Group II, III and IV elements are 2, 3 and 4 respectively; and that of Groups V, VI and VII are 3 (8 – 5), 2 (8 – 6) and 1 (8 – 7) respectively.

7 Why is no valency given for Group VIII?

In ionic compounds, the metal has a positive charge and the non-metal has a negative charge. (Remember, in ionic compounds there is a transfer of electrons between atoms; in covalent substances electrons are shared.)

The sum of the valencies of the atoms in a compound must be equal. This rule can be used to work out the formulae of compounds.

Consider silicon (Si, Group IV) and oxygen (O, Group IV). We can work out that silicon has a valency of 4 and oxygen has a valency of 2. Therefore, two oxygen atoms are needed to 'balance' one silicon atom and so the formula is SiO_2.

A different approach is to write the valency below each atom and cross multiply (see figure 1). This gives the formula Si_2O_4, which we reduce by the lowest common denominator (in this case, divide by 2) to give SiO_2.

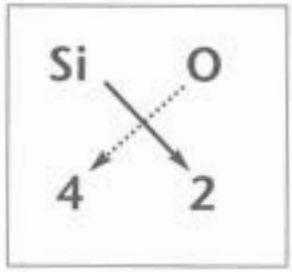


Figure 1

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8 Use the procedure shown in figure 1 to work out the formulae of the covalent compounds formed between these non-metals. Include the name of each compound.

- **a** phosphorus and chlorine
- **b** carbon and oxygen
- **c** boron and oxygen
- **d** sulfur and carbon
- **e** nitrogen and hydrogen
- **f** hydrogen and sulfur
- **g** silicon and fluorine

9 Give the general name for groups such as sulphate -SO_4^{2-}, carbonate -CO_3^{2-}, and nitrate -NO_3^{2-}.

10 Mark each statement as true or false.

- **a** A group of atoms that acts as a single unit in a reaction is called a radical.
- **b** When a metal combines with a non-metal the name of the compounds ends in '-ide'.
- **c** When a metal and a non-metal combine the non-metal is always named first in the compound.
- **d** Valency charges are always negative.
- **e** Valency gives an indication of the number or type (single, double, triple) of bond that can be formed in a compound.
- **f** In ionic compounds the metal has a positive charge and the non-metal has a negative charge.

11 State how many atoms of each substance are found in the simplest unit of each compound. The first one has been done for you.

	Compound	Formula	Type and number of atoms
	sodium silicate	Na_2SiO_3	2 sodium, 1 silicon, 3 oxygen
a	calcium hydroxide	$Ca(OH)_2$	
b	calcium phosphate	$Ca_3(PO4)_2$	
c	zinc nitrate	$Zn(NO_3)_2$	
d	sodium carbonate	Na_2CO_3	
e	potassium fluoride	KF	
f	copper sulfate	$CuSO_4$	
g	ammonium sulfate	$(NH_4)_2SO_4$	

Formulae of ionic compounds

Ionic compounds are formed between metals and non-metals. Remember, the overall charge on an ionic compound is zero, so the positive charges of metals must be balanced by an equal number of negative charges from non-metals.

12 Give the valency of:

- **a** Group I elements.
- **b** Group II elements.
- **c** Group III elements.

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Transition means the change from one stage to another. Transition elements are those found in the columns of the Periodic Table where the transition is made from the metals to non-metals (e.g. scandium to zinc, yttrium to cadmium, hafnium to mercury). Transition elements have variable valencies. The valency of a transition metal in a particular compound is usually shown by Roman numerals.

Copper (II) sulfate means that copper is present as Cu^{2+} in this compound; iron (III) oxide means that iron is present as Fe^{3+} in this compound.

EXAMPLES

i What is the formula of the compound formed between sodium (Group I metal) and chlorine (Group VII non-metal)?

Sodium forms the Na^+ ion and chlorine forms the Cl^- ion. The ratio must be 1:1, so the formula is NaCl.

ii What is the formula of the compound formed between magnesium (Group II metal) and oxygen (Group VI non-metal)?

Magnesium forms the Mg^{2+} ion and oxygen forms the O^{2-} ion. The ratio is 2:2; reduce by the lowest common multiple to 1:1. The formula is MgO.

iii What is the formula of iron (III) sulfate?

Iron (III) is Fe^{3+}. The sulfate radical has a formula of $-SO_4^{2-}$. In order to achieve neutral charge, two Fe^{3+} ions are needed to balance the charge on three SO_4^{2-} ions. (Or follow the cross multiplication method as shown in figure 2.) Hence the formula is $Fe_2(SO_4)_3$.

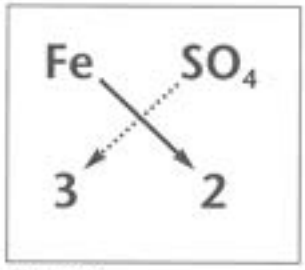


Figure 2

13 Copy and complete the following grids by writing the appropriate chemical formulae. The first one has been done for you.

Note that the ammonium radical (NH_4^+) is not a metal but behaves as one when writing formulae.

	Chlorine: chloride, Cl^-	Oxygen: oxide, O^{2-}	Sulfur: sulfide, S^{2-}	Nitrogen: nitride, N^{3-}
Sodium, Na^+	NaCl			
Magnesium, Mg^{2+}				
Iron (III), Fe^{3+}				
Ammonium, NH_4^+				

	Hydroxide: OH^-	Sulfate: SO_4^{2-}	Carbonate: CO_3^{2-}	Phosphate: PO_4^{3-}
Lithium, Li^+				
Calcium, Ca^{2+}				
Aluminium, Al^{3+}				
Ammonium, NH_4^+				

14 In the formula iron (II) oxide, what is signified by the Roman numerals?

15 'Metals in the middle of the Periodic Table, which have variable valencies.' What does statement define?

16 'A number assigned to an element that dictates how many atoms of a given element it can combine with.' What does statement define?

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